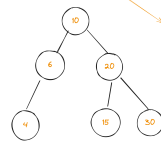


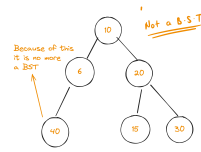
Binary Search Trees → BST



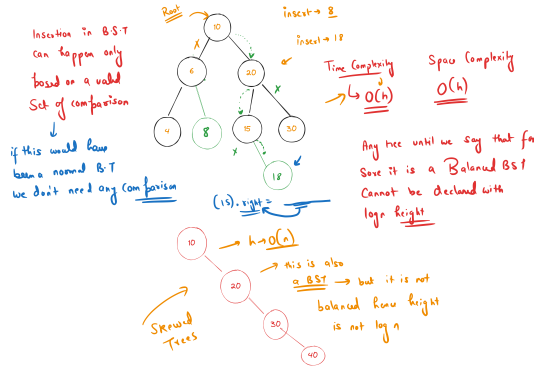
- optimise search a lot
- If you take inorder → it is sorted
- Balanced BST

4, 6, 15, 20, 30 → sorted

For every subtree we follow a rule i.e. all the elements in the left subtree should be having values lesser than root, and all the values in the right subtree should be having values greater than root.
And this fact should be recursively true for all subtrees



Insertion in a BST

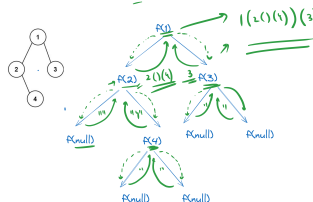
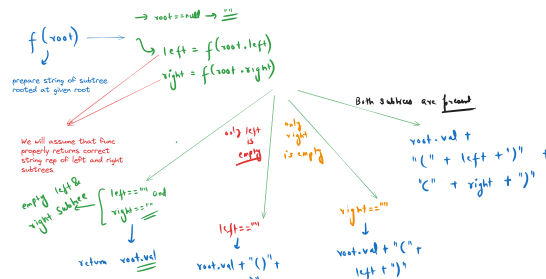


Leetcode 606



$1(2(1(1))(3))$

Preorder → Recursion



```

f(root) {
    if (root == null) return "";
    left = f(root.left);
    right = f(root.right);
    if (left == "" and right == "")
        return "" + root.val;
    else if (left == "")
        return root.val + "(" + right + ")";
    else if (right == "")
        return root.val + "(" + left + ")";
    else
        return root.val + "(" + left + ")" + "(" + right + ")";
}
    
```